Q13 — Discussion

DLS 2026 — AI Internals Module

Framing

The student question — *"How do models represent and store knowledge, and use it to answer?"* — cuts across several active research threads in mechanistic interpretability and knowledge editing. A useful first-order picture is that factual associations are stored, in substantial part, in the feed-forward (MLP) sublayers of transformers, which behave like key–value memories that can be probed and edited [1, 2]. Yet this picture is incomplete: knowledge is also distributed, encoded in attention routing, superposed across polysemantic directions, and in modern systems increasingly supplemented by non-parametric retrieval [3]. The questions below are deliberately open and are meant to structure our seminar discussion rather than to admit tidy answers.

Q1. Localization versus distribution. The ROME line of work suggests that specific facts can be traced to a small set of mid-layer MLP parameters via causal tracing [2], and that editing those parameters updates the stored association. But follow-up critiques argue that successful localization-based editing does not imply that knowledge is truly *stored* at the edited site, and that editing success and localization can dissociate [4]. *To what extent is a "fact" a localized object in weight space at all, versus an emergent property of distributed computation?* What experimental signatures would distinguish genuine storage from mere intervention sensitivity?

Q2. The representational substrate: features, directions, and superposition. If concepts correspond to directions in activation space, the model may pack many more features than it has neurons via *superposition*, trading interference for capacity [5]. Sparse autoencoders have recently been used to decompose activations into more monosemantic, interpretable features [6]. *Does "knowledge" live more naturally at the level of these recovered features than at the level of neurons or weights?* How should we reconcile a feature-level account with the parameter-level account from Q1, and what does superposition imply for the reliability of knowledge editing?

Q3. From storage to retrieval: the circuit that answers. Storing an association is distinct from *using* it to produce an answer. Work on circuit analysis — e.g., induction heads that implement in-context copying [7] and the tracing of *where* factual information is moved and surfaced at the final token [8] — suggests attention heads route subject information to the position where the prediction is read out. *What is the division of labor between MLP "lookup" and attention "routing" during factual recall, and does this division change with scale, fine-tuning, or chain-of-thought prompting?*

Q4. Parametric memory versus retrieval augmentation. Retrieval-augmented generation externalizes knowledge into a non-parametric store that is queried at inference time [3, 9]. This offers updatability and provenance, but raises the question of how retrieved context interacts with — and sometimes contradicts — parametric beliefs. *When internal parametric knowledge and retrieved evidence conflict, what governs which one the model trusts, and can we build*

mechanistic tests for this arbitration? What are the trade-offs between editing weights (Q1) and editing the corpus the model reads?

Q5. Faithfulness: does the model "know" what it uses? Even when a model answers correctly, its explanations or stated reasoning may not reflect the computation that produced the answer; chain-of-thought rationales can be unfaithful and post-hoc [10]. *How do we verify that an interpretability account of knowledge use is faithful to the underlying mechanism rather than a plausible narrative we impose?* What standards of causal evidence (ablation, patching, counterfactual editing) should the field adopt before claiming that it has explained how a model "knows" and "answers"?

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